

Grid-Aware Pricing for EV Charging Hubs in Smart Cities: A Dual-Layer Learning Approach to Manage Congestion and Flexibility

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Abstract—With the rapid electrification of urban mobility, public electric vehicle (EV) charging stations are evolving into grid-interactive hubs that couple transportation networks with local photovoltaic (PV) generation, battery storage, and distribution grids. Dynamic pricing for such hubs is challenging because charging demand depends jointly on prices, congestion-dependent travel times, and station waiting times, while users may instead select private, delayed, or cancelled charging according to unobserved reservation costs. Meanwhile, hub operators must compete for demand while respecting local energy-balance and grid-import limits. This paper develops a traffic- and grid-aware pricing framework for competing PV-storage charging hubs. EV charging choice is represented by a generalized-cost model that incorporates charging price, travel time, waiting time, and an outside option. The interaction among hubs is formulated as a generalized Nash equilibrium problem (GNEP), in which independently operated hubs select local prices under constraints coupled through user demand, queues, local energy availability, and distribution-grid capacity. A city-level coordinator aligns the decentralized policies with a shared aggregate-profit objective. To address incomplete information about user preferences, we propose a dual-layer multi-agent reinforcement learning (MARL) framework. The lower layer estimates latent outside-option preferences from realized charging demand, whereas the upper layer employs multi-agent proximal policy optimization (MAPPO) to learn decentralized pricing policies through centralized training. The reward and constraint-handling mechanisms jointly account for profitability, traffic congestion, queueing performance, and grid-capacity compliance under time-varying PV generation and battery states. The proposed framework provides a unified adaptive mechanism for coordinating competitive charging demand and local energy flexibility without requiring complete prior knowledge of user behavior.

Index Terms—EV charging pricing, photovoltaic-storage charging hubs, generalized Nash equilibrium, multi-agent reinforcement learning, traffic congestion, grid capacity, user preference inference.

I. INTRODUCTION

The rapid electrification of urban mobility is transforming public electric vehicle (EV) charging stations from passive electricity loads into grid-interactive charging hubs. By connecting transportation networks, distribution grids, and distributed energy resources, these hubs can support low-carbon mobility while providing demand-side flexibility to the power

system. This role becomes increasingly important in dense urban areas, where concentrated fast-charging demand may coincide with local load peaks and place substantial stress on distribution-network infrastructure.

Equipping charging hubs with photovoltaic (PV) generation and battery energy storage can partly alleviate this stress. Local PV energy can serve charging demand directly, while battery storage can shift energy across time and reduce exposure to peak grid electricity prices. However, the resulting operating environment is strongly time varying: PV output, battery state of charge, electricity procurement prices, and grid import capacity jointly determine the cost and amount of charging service that a hub can provide. Consequently, a retail charging price should reflect not only market demand but also the local energy flexibility available at the hub.

The spatial nature of public charging introduces an additional layer of coupling. EV users compare charging hubs according to more than the announced price; congestion-dependent travel time, detour inconvenience, and expected station waiting time also affect their decisions. A low price can attract demand and improve the use of local renewable energy, but it can also lengthen queues, exceed local service or grid capacity, and draw traffic through congested road segments. A high price can relieve these pressures, yet may leave chargers and available PV-storage resources underutilized. Pricing must therefore coordinate mobility-side demand responses with station- and energy-side operating conditions.

This coordination problem is further complicated by market competition and incomplete information. Public charging hubs are commonly operated by independent profit-seeking entities, and the price announced by one hub changes the demand, queues, and operating conditions faced by its competitors. Moreover, operators do not directly observe users' willingness to accept public charging. In particular, the generalized cost of the outside option such as home charging, delayed charging, or trip cancellation—is private and determines whether a user enters the public charging market at all. These features motivate a traffic- and grid-aware pricing framework that represents strategic competition while learning latent user preferences

from observed demand.

A. Related Work and Research Gap

Dynamic pricing for EV charging has been widely studied as a means of shifting flexible demand, reducing electricity procurement costs, and improving charging station profitability. Most energy-oriented formulations determine retail prices from wholesale electricity prices, anticipated charging demand, or distribution-network loading. Although these approaches capture the temporal flexibility of EV charging, users' spatial station choices and the mobility costs associated with reaching a station are often simplified. A price that is effective from an energy perspective may therefore concentrate demand at a particular hub or redirect users through congested parts of the road network.

A related body of work couples transportation networks with EV charging infrastructure. Traffic-aware charging models account for the spatial distribution of charging demand and incorporate travel time, detour cost, or station congestion into charging decisions. Many such studies, however, adopt the viewpoint of a central system planner that directly controls charging flows or prices to minimize total travel time, system cost, or social cost. This setting does not capture decentralized price decisions and the coupled operating constraints faced by independently operated charging hubs.

Research on PV-storage-integrated charging hubs has primarily focused on energy management, including the scheduling of PV generation, battery charging and discharging, and grid electricity purchases. These studies demonstrate the value of local renewable energy and storage for reducing grid dependence and operating costs. Nevertheless, energy dispatch and retail pricing are often treated separately, and the effect of time-varying local energy availability on competition for spatially distributed charging demand remains insufficiently represented.

Game-theoretic pricing models provide a natural way to describe competition among charging operators. In a conventional Nash game, each operator selects a price to maximize its own payoff while demand is redistributed according to the relative attractiveness of competing stations. In the setting considered here, however, the prices of all hubs jointly determine each hub's demand and, through that demand, its waiting time, charging-energy requirement, and grid import. The feasible pricing set of one operator consequently depends on the prices selected by the others. This endogenous coupling calls for a generalized Nash equilibrium problem (GNEP), rather than a game with independent feasible sets.

Reinforcement learning (RL) and multi-agent reinforcement learning (MARL) have also been applied to charging control and competitive pricing. These methods can adapt policies through repeated interaction in time-varying and strategically coupled environments. Yet many learning-based pricing models assume that the user-response function and its behavioral parameters are known or directly observable. This assumption is restrictive when users can reject public charging in favor of an outside option whose cost is private. Ignoring this option

can systematically overpredict public charging demand and distort both pricing and feasibility assessments.

Taken together, the literature leaves a gap at the intersection of four features: congestion-dependent charging-hub choice, decentralized pricing, PV-storage and grid constraints, and unknown outside-option preferences. Existing studies generally address only a subset of these features or rely on complete knowledge of user behavior. This paper addresses the combined problem by linking a traffic-energy model and a shared-constraint, welfare-aligned GNEP with a dual-layer learning framework that updates user-preference estimates and pricing policies jointly.

B. Problem Statement and Main Contributions

We consider multiple independently operated PV-storage-integrated charging hubs serving spatially distributed EV demand. At each decision period, every hub selects its own retail charging price from local information. The joint price vector affects users' hub and outside-option choices, which in turn determine station demand, waiting time, charging-energy requirements, and local grid imports. Each operator must account for price bounds, service-quality limits, local energy balance, battery operating limits, and grid import capacity. To coordinate the resulting cross-hub externalities, a city-level coordinator trains the decentralized pricing policies using weighted aggregate hub profit, while each hub retains independent local execution. At the same time, the outside-option cost governing users' willingness to use public charging is not known a priori and must be inferred from realized demand. The central question is therefore how independently operated hubs can learn decentralized, high-welfare pricing policies while accounting for traffic congestion, station queues, and energy constraints under incomplete user information.

The main contributions of this paper are summarized as follows:

- **An integrated traffic-grid-energy pricing model:** We develop a dynamic model that links congestion-dependent travel costs, charging-hub choice with an explicit outside option, station-level queueing, and PV-storage-grid operation. The model makes the demand and profitability of each charging hub responsive to both mobility conditions and local energy availability.
- **A welfare-aligned GNEP formulation for charging hubs:** We formulate the interaction among independently operated hubs as a shared-constraint, welfare-aligned GNEP. The joint prices redistribute charging demand and thereby couple the hubs' feasible strategy sets through waiting-time and full-delivery requirements. The hubs retain local price decisions, while a common weighted aggregate-profit objective selects a welfare-optimal GNE. This formulation distinguishes the proposed model from centralized price execution and standard Nash pricing games with independent feasible sets.
- **A dual-layer MARL framework under unknown user preferences:** We propose a lower-layer inference mechanism that updates the latent outside-option cost from

the discrepancy between predicted and realized charging demand. An upper-layer multi-agent proximal policy optimization (MAPPO) method then learns decentralized continuous pricing policies using centralized training. Its shared reward combines weighted aggregate hub profit with penalties for waiting-time excess and unmet energy, enabling the agents to adapt to traffic conditions, PV output, battery states, grid limits, and inferred user preferences.

C. Paper Organization

The remainder of this paper is organized as follows. Section II presents the traffic network, charging-hub choice, station queueing, PV-storage-grid operation, and profit models. Section III formulates the welfare-aligned, shared-constraint pricing problem as a GNEP and explains the difficulty created by unknown user preferences. Section IV develops the dual-layer preference-inference and MAPPO-based pricing framework. Section V presents the numerical case studies, and Section VI concludes the paper.

II. SYSTEM MODEL

We consider a smart-city charging ecosystem that couples an urban transportation network, a set of independently operated EV charging hubs, and their local energy systems. The transportation network is represented by a directed graph $\mathcal{G}^T = (\mathcal{V}^T, \mathcal{E}^T)$. A set of origin-destination (OD) pairs $\mathcal{K}$ generates potential EV charging requests, and a set of hubs $\mathcal{I} = \{1, \dots, I\}$ provides public charging services. Each hub is equipped with multiple chargers, photovoltaic (PV) generation, battery energy storage, and a grid connection with limited import capacity. The interaction among these components is summarized in Fig. 1.

Time is divided into decision periods of duration Δt . At the beginning of period t , the road traffic state, potential charging demand, PV availability, battery state, and grid electricity price are observed. Each hub $i \in \mathcal{I}$ then announces a retail charging price $p_i(t)$. EV users compare the generalized costs of the available hubs and an outside option, such as home charging, delayed charging, or trip cancellation. The resulting hub choices determine station arrivals and requested charging energy. After demand is realized, each hub updates its queueing condition and applies a local PV-storage-grid dispatch policy. These outcomes determine the hub's operating profit and the state observed in the next period.

This timing convention separates information available when users choose a hub from outcomes realized after their choices. In particular, users respond to an expected waiting time known at the beginning of period t , whereas the actual waiting time in period t is computed after demand is allocated. This avoids an algebraic loop between same-period hub choice and queueing delay and is consistent with the state-transition structure used by the learning framework in Section IV.

A. Traffic Network and Congestion-Dependent Travel Time

For each OD pair $k \in \mathcal{K}$, let $D_k(t)$ denote the number of potential EV charging requests generated during period t .

Thus, $D_k(t)$ is charging demand before users choose a public hub or the outside option; it is not the total traffic demand on the road network. Let $x_e(t)$ denote the observed aggregate traffic flow on road link $e \in \mathcal{E}^T$, including background traffic. The corresponding travel time is described by the Bureau of Public Roads (BPR) function

$$T_e(t) = T_e^0 \left[1 + a \left(\frac{x_e(t)}{C_e} \right)^b \right], \quad (1)$$

where T_e^0 is the free-flow travel time, C_e is the link capacity, and $a, b > 0$ are congestion parameters.

Let $r_{k,i}$ be the candidate access route from OD pair k to hub i , including any detour required to reach the hub. The congestion-dependent travel time associated with this choice is

$$T_{k,i}^{\text{road}}(t) = \sum_{e \in r_{k,i}} T_e(t). \quad (2)$$

The access routes are treated as fixed within a decision period. Moreover, $x_e(t)$ is an exogenous but observable traffic state: the pricing agents respond to road congestion but do not directly control network flows or optimize total travel time. Station congestion, by contrast, is endogenous because hub prices redistribute charging arrivals. This distinction will also be maintained when interpreting the numerical results.

B. EV Charging-Hub Choice with an Outside Option

Before making a choice in period t , users observe or receive an estimate of the waiting time at each hub. Denote this information by $\tilde{T}_i^{\text{wait}}(t)$. A directly implementable baseline is the most recently realized waiting time,

$$\tilde{T}_i^{\text{wait}}(t) = T_i^{\text{wait}}(t-1), \quad (3)$$

although a short-term queue forecast can be used without changing the model.

For a user from OD pair k , the generalized monetary cost of choosing hub i is

$$C_{k,i}(t) = p_i(t)E_k^{\text{ch}} + \nu \left(T_{k,i}^{\text{road}}(t) + \tilde{T}_i^{\text{wait}}(t) \right), \quad (4)$$

where E_k^{ch} is the average requested charging energy and ν is the value of time. Hence, all terms in $C_{k,i}(t)$ are expressed in monetary units.

Users may instead choose an outside option, indexed by 0. Its generalized cost C_k^{out} represents the reservation cost associated with alternatives such as home charging, delayed charging, or trip cancellation. This cost is private information and is not directly observed by the hub operators. Using a multinomial logit model, the probability that a user from OD pair k chooses hub i is

$$P_{k,i}(t) = \frac{\exp[-\lambda C_{k,i}(t)]}{\exp[-\lambda C_k^{\text{out}}] + \sum_{j \in \mathcal{I}} \exp[-\lambda C_{k,j}(t)]}, \quad i \in \mathcal{I}, \quad (5)$$

and the outside-option probability is

$$P_{k,0}(t) = \frac{\exp[-\lambda C_k^{\text{out}}]}{\exp[-\lambda C_k^{\text{out}}] + \sum_{j \in \mathcal{I}} \exp[-\lambda C_{k,j}(t)]}, \quad (6)$$

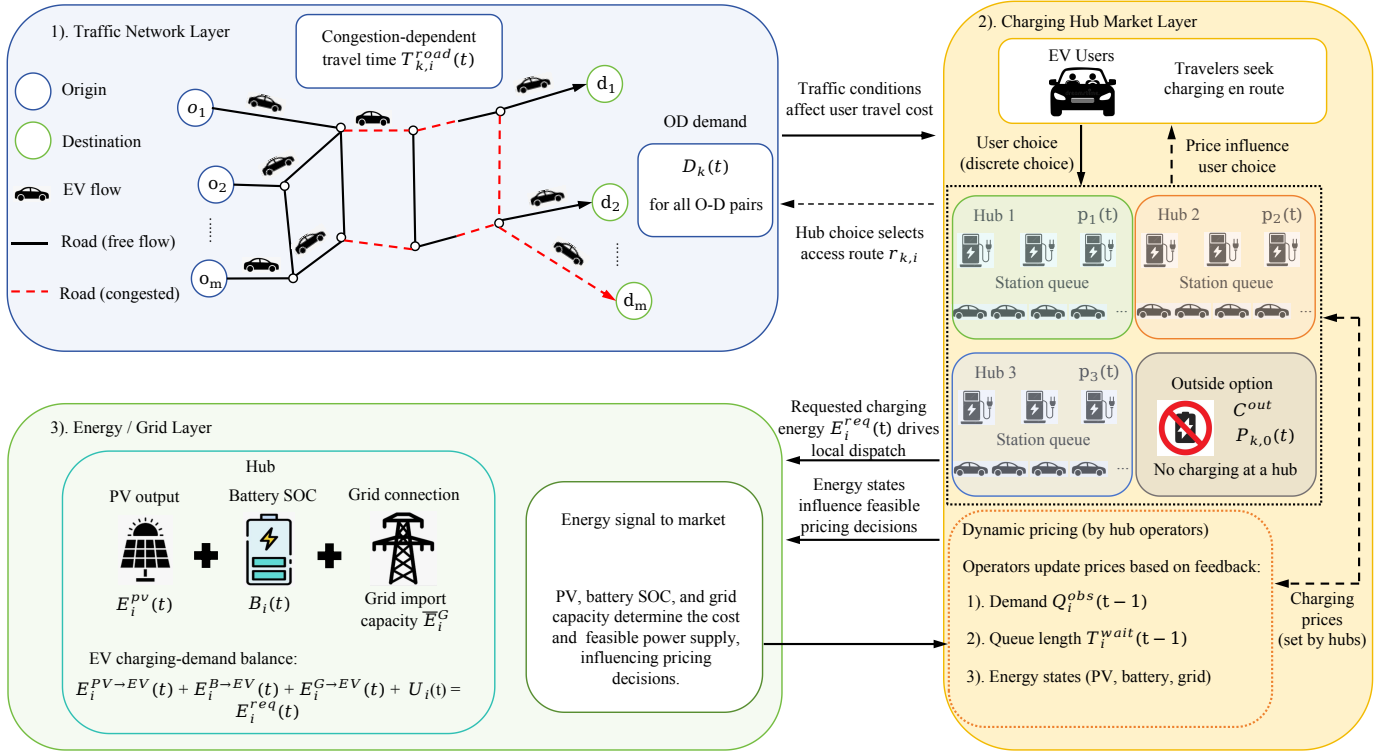


Fig. 1. Traffic-energy-market interaction in the proposed EV charging hub network. Charging hubs announce prices to influence EV route-and-charge choices, while realized demand feeds back to station queueing conditions, local PV-storage dispatch, and grid import requirements.

where $\lambda > 0$ is the inverse-cost sensitivity. The probabilities satisfy $P_{k,0}(t) + \sum_{i \in \mathcal{I}} P_{k,i}(t) = 1$.

The OD-specific costs $\{C_k^{out}\}_{k \in \mathcal{K}}$ can be estimated separately only when demand observations retain OD information. If the operators observe only aggregate hub-level counts, we impose the shared specification $C_k^{out} = C^{out}$ for all $k \in \mathcal{K}$. This is the identifiable baseline used by the lower-layer inference model in Section IV; OD-specific inference is treated as an extension for richer data.

The mean number of charging requests allocated to hub i is

$$Q_i(t) = \sum_{k \in \mathcal{K}} D_k(t) P_{k,i}(t), \quad (7)$$

and $\mathbf{Q}(t) = [Q_1(t), \dots, Q_I(t)]^\top$. The observed count, denoted by $Q_i^{obs}(t)$, may differ from this conditional mean because of finite-population sampling and measurement noise. A deterministic simulation may set $Q_i^{obs}(t) = Q_i(t)$, whereas the preference-inference layer in Section IV uses the observed quantity. This distinction prevents “predicted” and “realized” demand from being used interchangeably.

The requested charging energy at hub i is

$$E_i^{req}(t) = \sum_{k \in \mathcal{K}} E_k^{ch} D_k(t) P_{k,i}(t). \quad (8)$$

Both $Q_i(t)$ and $E_i^{req}(t)$ depend on the full price vector, traffic conditions, expected waiting times, and the latent outside-option cost.

C. Station-Level Queueing Model

Let N_i be the number of chargers at hub i and τ_i the average service time per EV. The approximate number of EVs that can begin service during one decision period is

$$S_i = \frac{N_i \Delta t}{\tau_i}. \quad (9)$$

Given the allocated arrivals, the realized waiting time is approximated by

$$T_i^{wait}(t) = \tau_i \left[\frac{Q_i(t)}{S_i} - 1 \right]^+, \quad [z]^+ = \max\{0, z\}. \quad (10)$$

This deterministic approximation is intended to retain the main congestion effect without adding a detailed stochastic queue to the learning environment. When $Q_i(t) \leq S_i$, waiting is negligible; above this threshold, waiting increases with the overload ratio. The resulting value becomes observable after period t and supplies the expected waiting-time input in Eq. (3) for the next period.

Each hub is subject to the quality-of-service requirement

$$T_i^{wait}(t) \leq T_i^{\max}, \quad i \in \mathcal{I}. \quad (11)$$

Because $Q_i(t)$ depends on all announced prices, the induced waiting-time condition is strategically coupled across hubs. This is the endogenous congestion mechanism used in the GNEP and the upper-layer reward.

D. PV-Storage-Grid Operation

All power quantities are converted to energy over one decision period. Let $E_i^{\text{PV}}(t)$ denote available PV energy and $\bar{E}_i^{\text{G}} = \bar{P}_i^{\text{G}} \Delta t$ the maximum grid energy that can be imported during period t . The PV allocation satisfies

$$E_i^{\text{PV}}(t) = E_i^{\text{PV} \rightarrow \text{EV}}(t) + E_i^{\text{PV} \rightarrow \text{B}}(t) + E_i^{\text{PV}, \text{cur}}(t), \quad (12)$$

with all terms on the right-hand side nonnegative. Unless explicitly stated otherwise, all energy-flow, delivered-energy, and unmet-energy variables in this subsection are nonnegative.

To distinguish requested service from physically delivered service, define $E_i^{\text{srv}}(t)$ as the energy delivered to EVs and $U_i(t)$ as unmet or deferred charging energy. They satisfy

$$E_i^{\text{req}}(t) = E_i^{\text{srv}}(t) + U_i(t), \quad U_i(t) \geq 0, \quad (13)$$

$$E_i^{\text{srv}}(t) = E_i^{\text{PV} \rightarrow \text{EV}}(t) + E_i^{\text{B} \rightarrow \text{EV}}(t) + E_i^{\text{G} \rightarrow \text{EV}}(t). \quad (14)$$

For a feasible price in the hard-constrained game, $U_i(t) = 0$. Retaining $U_i(t)$ explicitly also allows the learning environment to evaluate and penalize temporary energy shortfalls during policy exploration.

Let $B_i(t)$ denote the battery energy content, measured in kWh, with capacity $\bar{B}_i$. The normalized state of charge reported to the learning agent is $\text{SOC}_i(t) = B_i(t)/\bar{B}_i$. Battery dynamics are

$$B_i(t+1) = B_i(t) + \eta_i^{\text{ch}} (E_i^{\text{PV} \rightarrow \text{B}}(t) + E_i^{\text{G} \rightarrow \text{B}}(t)) - \frac{1}{\eta_i^{\text{dis}}} E_i^{\text{B} \rightarrow \text{EV}}(t), \quad (15)$$

subject to

$$\underline{B}_i \leq B_i(t+1) \leq \bar{B}_i, \quad (16)$$

$$0 \leq E_i^{\text{PV} \rightarrow \text{B}}(t) + E_i^{\text{G} \rightarrow \text{B}}(t) \leq \bar{P}_i^{\text{ch}} \Delta t, \quad (17)$$

$$0 \leq E_i^{\text{B} \rightarrow \text{EV}}(t) \leq \bar{P}_i^{\text{dis}} \Delta t. \quad (18)$$

Simultaneous charging and discharging is excluded by

$$(E_i^{\text{PV} \rightarrow \text{B}}(t) + E_i^{\text{G} \rightarrow \text{B}}(t)) E_i^{\text{B} \rightarrow \text{EV}}(t) = 0. \quad (19)$$

The grid connection constraint is

$$0 \leq E_i^{\text{G} \rightarrow \text{EV}}(t) + E_i^{\text{G} \rightarrow \text{B}}(t) \leq \bar{E}_i^{\text{G}}. \quad (20)$$

The hub operator's strategic action remains the retail price. Once charging demand is realized, the energy variables are produced by an embedded dispatch module rather than selected by the pricing agent. Let $e_i(t)$ collect the variables in Eqs. (12)–(20), and let $\mathcal{E}_i(t)$ denote their physical feasible set. The dispatch is defined by

$$e_i^*(t) \in \arg \min_{e_i(t) \in \mathcal{E}_i(t)} \{C_i^{\text{energy}}(t) + \kappa_i^U U_i(t) - v_i(t) B_i(t+1)\}, \quad (21)$$

where κ_i^U is chosen sufficiently large to prioritize serving charging requests and $v_i(t)$ represents the estimated continuation value of stored energy. The latter allows surplus PV or

low-price grid energy to be stored when its expected future value justifies doing so. The actual energy cost is

$$C_i^{\text{energy}}(t) = \pi^{\text{G}}(t) (E_i^{\text{G} \rightarrow \text{EV}}(t) + E_i^{\text{G} \rightarrow \text{B}}(t)) + c_i^{\text{B}} E_i^{\text{B} \rightarrow \text{EV}}(t) + c_i^{\text{PV}} (E_i^{\text{PV} \rightarrow \text{EV}}(t) + E_i^{\text{PV} \rightarrow \text{B}}(t)), \quad (22)$$

where $\pi^{\text{G}}(t)$ is the grid electricity price, c_i^{B} is the battery degradation cost per unit discharged, and c_i^{PV} is the PV operation and maintenance cost. The terms $\kappa_i^U U_i(t)$ and $-v_i(t) B_i(t+1)$ guide the dispatch but are not treated as cash expenses in the profit function.

Equation (21) defines a deterministic mapping from the requested energy and local energy state to dispatch, delivered energy, unmet energy, and next-period battery state; a fixed tie-breaking rule is applied if the optimizer is not unique. It therefore closes the model needed by Sections III and IV while preserving price as the only strategic and learning action.

E. Charging Hub Profit Model

Because energy shortages may occur during policy exploration, revenue is based on delivered rather than requested energy:

$$R_i(t) = p_i(t) E_i^{\text{srv}}(t). \quad (23)$$

For the service-related operating cost, we use a proportional-service approximation. Define

$$\alpha_i(t) = \begin{cases} E_i^{\text{srv}}(t)/E_i^{\text{req}}(t), & E_i^{\text{req}}(t) > 0, \\ 1, & E_i^{\text{req}}(t) = 0, \end{cases} \quad (24)$$

$$Q_i^{\text{srv}}(t) = \alpha_i(t) Q_i(t).$$

The service operating cost is

$$C_i^{\text{op}}(t) = c_i^{\text{op}} Q_i^{\text{srv}}(t), \quad (25)$$

where c_i^{op} is the cost per served charging transaction.

The one-period operating profit is then

$$\Pi_i(t) = p_i(t) E_i^{\text{srv}}(t) - C_i^{\text{energy}}(t) - c_i^{\text{op}} Q_i^{\text{srv}}(t). \quad (26)$$

Under hard feasibility, $U_i(t) = 0$, so all requested energy is served and the profit reduces to the standard revenue-minus-cost expression. During MARL exploration, the distinction between requested and served quantities prevents a hub from earning revenue on energy that it cannot physically deliver.

The profit in Eq. (26) depends on the full price vector. The prices jointly determine users' hub choices, while the induced charging demand determines queueing, dispatch, and energy cost. Hence competitors' prices affect both the payoff and feasibility of hub i , providing the coupling used in the game-theoretic formulation of Section III.

III. WELFARE-ALIGNED GENERALIZED NASH PRICING FORMULATION

This section formulates decentralized charging-hub pricing as a generalized Nash equilibrium problem (GNEP). At each decision period, all hubs choose their prices simultaneously from local information and do not observe the contemporaneous prices or private states of the other hubs. Their

decisions are nevertheless coupled because the joint price vector determines user allocation, station queueing, and the energy demand passed to each hub's embedded dispatch. Consequently, the set of prices that is feasible for one hub depends on the prices selected by the others.

Unlike a purely competitive GNEP in which each hub maximizes only its own profit, the formulation adopts a welfare-aligned objective. A city-level charging coordinator uses the aggregate performance of all hubs when training their decentralized pricing policies. The resulting identical-interest GNEP retains the coupled feasible sets required by the system model while selecting prices according to system-wide benefit.

A. Players, Local Information, and Simultaneous Decisions

The players are the charging hubs indexed by $\mathcal{I} = \{1, \dots, I\}$. At the beginning of period t , hub i observes

$$o_i(t) = (Q_i^{\text{obs}}(t-1), p_i(t-1), T_i^{\text{wait}}(t-1), E_i^{\text{PV}}(t), B_i(t), \pi^G(t), \bar{E}_i^G, \varphi_i^{\text{tr}}(t), \hat{C}^{\text{out}}(t)), \quad (27)$$

where $\varphi_i^{\text{tr}}(t)$ summarizes locally available traffic information and $\hat{C}^{\text{out}}(t)$ is supplied by the lower preference-inference layer. Importantly, $o_i(t)$ does not include any current price $p_j(t)$ or private observation $o_j(t)$ for $j \neq i$.

Hub i selects the retail price

$$a_i(t) = p_i(t) \in \mathcal{A}_i = [p_i^{\min}, p_i^{\max}]. \quad (28)$$

All hubs act simultaneously. Their joint price vector is

$$\mathbf{p}(t) = [p_1(t), \dots, p_I(t)]^\top, \quad (29)$$

and $\mathbf{p}_{-i}(t)$ denotes the vector of all prices except $p_i(t)$. This notation describes strategic coupling and counterfactual deviations; it does not imply that hub i observes $\mathbf{p}_{-i}(t)$ when choosing its price.

For a fixed period, collect the observable physical and market variables in $\mathbf{z}(t)$. Given the joint price, state, and true outside-option cost, the system response is

$$\mathcal{R}(\mathbf{p}(t); \mathbf{z}(t), C^{\text{out}}) = \{Q_i(t), E_i^{\text{req}}(t), T_i^{\text{wait}}(t), \mathbf{e}_i^*(t), E_i^{\text{stpv}}(t), U_i(t), B_i(t+1), \Pi_i(t)\}_{i \in \mathcal{I}}. \quad (30)$$

The components of this mapping are defined by Eqs. (5)–(26). In particular, $\mathbf{e}_i^*(t)$ is generated by the embedded dispatch in Eq. (21); it is not an additional strategic action.

B. Coupled Feasible Pricing Sets

The embedded dispatch always satisfies the physical PV, battery, and grid constraints in Eqs. (12)–(20). Insufficient local energy or grid capacity therefore appears as unmet charging energy $U_i(t)$ rather than as a violation of those physical constraints. To align decentralized decisions with system-wide feasibility, we use a shared-constraint GNEP. For fixed $\mathbf{z}(t)$ and C^{out} , the feasible unilateral prices available to hub i are

$$\mathcal{S}_i^t(\mathbf{p}_{-i}) = \{p_i \in \mathcal{A}_i \mid T_j^{\text{wait}}(p_i, \mathbf{p}_{-i}) \leq T_j^{\text{max}}, U_j(p_i, \mathbf{p}_{-i}) = 0, \forall j \in \mathcal{I}\}. \quad (31)$$

The conditioning on $\mathbf{z}(t)$ and C^{out} is suppressed inside the response terms for readability.

The dependence of $\mathcal{S}_i^t$ on $\mathbf{p}_{-i}$ is the source of the generalized Nash structure. The joint prices alter the choice probabilities in Eq. (5), which changes $Q_j(t)$, $E_j^{\text{req}}(t)$, $T_j^{\text{wait}}(t)$, and $U_j(t)$ across the hub network. Thus, although each hub controls only its own price, its feasible price set is the corresponding slice of a shared system-feasible set and is not independent of the other hubs' decisions.

The joint feasible set is

$$\mathcal{F}^t = \left\{ \mathbf{p} \in \prod_{i \in \mathcal{I}} \mathcal{A}_i \mid p_i \in \mathcal{S}_i^t(\mathbf{p}_{-i}), \forall i \in \mathcal{I} \right\}. \quad (32)$$

We assume that $\mathcal{F}^t$ is nonempty for the operating states considered. If it is empty, pricing alone cannot meet all service requirements and an additional intervention, such as admission control or capacity expansion, is required.

C. Welfare-Aligned GNEP and Equilibrium Selection

Define the system-wide economic benefit as the weighted aggregate hub profit

$$W^t(\mathbf{p}) = \sum_{i \in \mathcal{I}} w_i \Pi_i(\mathbf{p}; \mathbf{z}(t), C^{\text{out}}), \quad w_i > 0. \quad (33)$$

Setting $w_i = 1$ gives total charging-hub profit. Other positive weights may be used to balance hubs of different scales.

In the welfare-aligned GNEP, every player evaluates its price using the same system objective. Given the other prices, hub i is associated with

$$\begin{aligned} \max_{p_i} \quad & W^t(p_i, \mathbf{p}_{-i}) \\ \text{s.t.} \quad & p_i \in \mathcal{S}_i^t(\mathbf{p}_{-i}). \end{aligned} \quad (34)$$

This best-response expression is a mathematical equilibrium condition, not the online decision rule used by a hub. During execution, prices are generated simultaneously from local policies as described in Section IV.

A jointly feasible vector $\mathbf{p}^{\text{GNE}}(t)$ is a GNE if, for every $i \in \mathcal{I}$,

$$p_i^{\text{GNE}}(t) \in \mathcal{S}_i^t(\mathbf{p}_{-i}^{\text{GNE}}(t)), \quad (35)$$

and

$$\begin{aligned} W^t(p_i^{\text{GNE}}, \mathbf{p}_{-i}^{\text{GNE}}) &\geq W^t(q_i, \mathbf{p}_{-i}^{\text{GNE}}), \\ \forall q_i &\in \mathcal{S}_i^t(\mathbf{p}_{-i}^{\text{GNE}}). \end{aligned} \quad (36)$$

Hence no hub can unilaterally select another feasible price that improves the shared system objective.

An arbitrary GNE need not be globally welfare maximizing. The desired equilibrium is therefore selected by

$$\mathbf{p}^*(t) \in \arg \max_{\mathbf{p} \in \mathcal{F}^t} W^t(\mathbf{p}). \quad (37)$$

Every global solution of Eq. (37) is a GNE because a profitable unilateral feasible deviation would contradict global optimality. The converse is not generally true. Equation (37) therefore identifies the welfare-optimal GNE targeted by the learning framework, rather than accepting any equilibrium of the generalized game.

D. Unknown Preferences and Decentralized Policy Learning

The welfare-aligned GNEP above is defined using the true outside-option cost. In practice, this parameter is private and the hubs observe only $Q_i^{\text{obs}}(t)$ after implementing their prices. Under aggregate demand observations, the lower layer estimates the shared $\hat{C}^{\text{out}}(t)$ and provides it as a common behavioral signal to the pricing actors.

Each actor uses a decentralized stochastic policy

$$p_i(t) \sim \pi_{\theta_i}(\cdot | o_i(t)), \quad \boldsymbol{\pi}_{\theta}(\mathbf{p} | \mathbf{o}) = \prod_{i \in \mathcal{I}} \pi_{\theta_i}(p_i | o_i). \quad (38)$$

This factorization ensures that no actor receives another hub's current price. During centralized training, however, a critic may use global trajectories to evaluate the shared system objective and coordinate policy updates.

To provide learning feedback when exploratory prices temporarily violate the waiting-time or full-delivery requirements, the hard welfare objective is represented during training by the shared reward

$$r^{\text{sys}}(t) = W^t(\mathbf{p}(t)) - \rho^{\text{wait}} \sum_{i \in \mathcal{I}} [T_i^{\text{wait}}(t) - T_i^{\text{max}}]^+ - \rho^{\text{unmet}} \sum_{i \in \mathcal{I}} U_i(t). \quad (39)$$

The PV, battery, and grid constraints remain hard enforced by the embedded dispatch. Section IV develops the dual-layer MAPPO method used to learn the mapping from local observations to a high-welfare, approximately feasible GNE pricing policy under the unknown user preference.

IV. PROPOSED DUAL-LAYER MAPPO FRAMEWORK

The welfare-aligned GNEP in Section III cannot be implemented directly because the outside-option cost is unknown and the traffic-energy state changes over time. We therefore develop a dual-layer learning framework. The lower layer estimates the shared latent outside-option cost from aggregate demand observations. The upper layer uses multi-agent proximal policy optimization (MAPPO) to learn decentralized pricing policies under centralized training.

At each period, every actor observes only its local information and selects a price simultaneously, as specified in Eq. (38). The centralized critic is used only during training and evaluates the shared system reward in Eq. (39). Thus, the learning architecture preserves decentralized execution while coordinating the actor updates toward the welfare-optimal GNE objective. The overall interaction among the physical environment, the lower-layer preference estimator, and the upper-layer actor-critic architecture is illustrated in Fig. 2.

A. Lower Layer: Shared User-Preference Inference

Because the available observations are aggregate hub-level counts, the main implementation uses the identifiable shared specification $C_k^{\text{out}} = C^{\text{out}}$ for all $k \in \mathcal{K}$, as discussed in Section II-B. Let $\hat{C}^{\text{out}}(t)$ denote its estimate before prices are selected in period t . Given the joint price, the traffic

state, and the expected waiting times, the estimated hub-choice probabilities are

$$\hat{P}_{k,i}(t) = \frac{\exp[-\lambda C_{k,i}(t)]}{\exp[-\lambda \hat{C}^{\text{out}}(t)] + \sum_{j \in \mathcal{I}} \exp[-\lambda C_{k,j}(t)]}, \quad (40)$$

and

$$\hat{P}_{k,0}(t) = \frac{\exp[-\lambda \hat{C}^{\text{out}}(t)]}{\exp[-\lambda \hat{C}^{\text{out}}(t)] + \sum_{j \in \mathcal{I}} \exp[-\lambda C_{k,j}(t)]}. \quad (41)$$

The predicted hub demand is

$$\hat{Q}_i(t) = \sum_{k \in \mathcal{K}} D_k(t) \hat{P}_{k,i}(t). \quad (42)$$

After the joint prices are implemented, the hubs observe $Q_i^{\text{obs}}(t)$. The instantaneous inference loss is

$$\mathcal{L}_{\text{inf}}(t) = \frac{1}{2I} \sum_{i \in \mathcal{I}} (Q_i^{\text{obs}}(t) - \hat{Q}_i(t))^2. \quad (43)$$

The relevant derivatives have closed forms:

$$\frac{\partial \hat{P}_{k,i}(t)}{\partial \hat{C}^{\text{out}}(t)} = \lambda \hat{P}_{k,i}(t) \hat{P}_{k,0}(t), \quad (44)$$

$$\frac{\partial \hat{Q}_i(t)}{\partial \hat{C}^{\text{out}}(t)} = \lambda \sum_{k \in \mathcal{K}} D_k(t) \hat{P}_{k,i}(t) \hat{P}_{k,0}(t). \quad (45)$$

Consequently,

$$\frac{\partial \mathcal{L}_{\text{inf}}(t)}{\partial \hat{C}^{\text{out}}(t)} = -\frac{1}{I} \sum_{i \in \mathcal{I}} (Q_i^{\text{obs}}(t) - \hat{Q}_i(t)) \frac{\partial \hat{Q}_i(t)}{\partial \hat{C}^{\text{out}}(t)}. \quad (46)$$

The estimate is updated by projected gradient descent:

$$\hat{C}^{\text{out}}(t+1) = \text{Proj}_{[C^{\text{min}}, C^{\text{max}}]} \left[\hat{C}^{\text{out}}(t) - \eta_C \frac{\partial \mathcal{L}_{\text{inf}}(t)}{\partial \hat{C}^{\text{out}}(t)} \right], \quad (47)$$

where $\eta_C > 0$ is the inference step size and the projection prevents behaviorally implausible estimates. If demand observations are noisy, the loss and gradient can be averaged over a short rolling window. The preference estimate is updated at every period, whereas the MAPPO parameters are updated after a rollout; this two-timescale implementation allows the actors to learn from a comparatively current demand model.

If predicted public demand is below observed demand, the gradient update tends to increase $\hat{C}^{\text{out}}$, making the outside option less attractive. The opposite adjustment occurs when predicted demand is too high. Persistent variation in prices and demand is required for reliable identification, and this condition is examined in the case study.

B. Upper Layer: Welfare-Aligned MAPPO

Actor i receives the local observation $o_i(t)$ defined in Eq. (27); in particular, it does not receive any other hub's current price. Let

$$\mathbf{s}^g(t) = [o_1(t), \dots, o_I(t)] \quad (48)$$

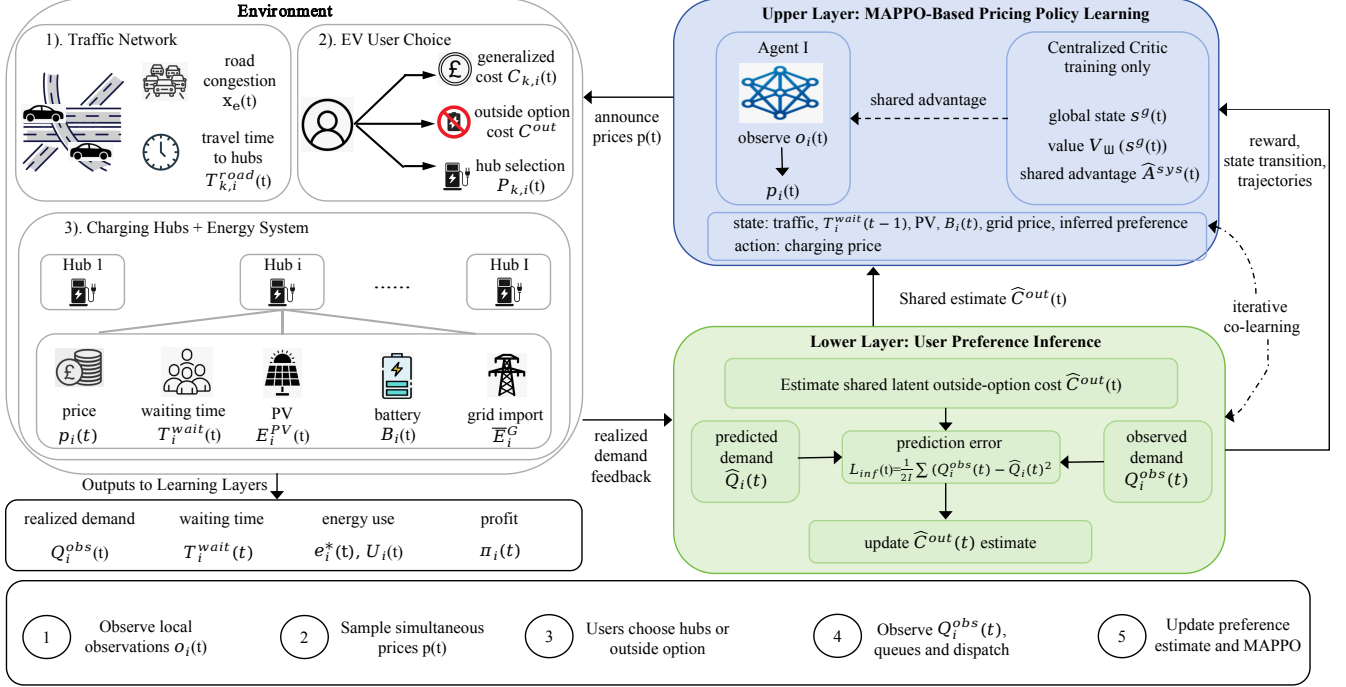


Fig. 2. Architecture of the proposed dual-layer MAPPO framework. The physical environment maps simultaneous hub prices to user choices, station queues, and embedded energy dispatch. The lower layer infers the shared outside-option cost from predicted and observed demand, while each decentralized actor selects a price from its local observation and the centralized critic uses the shared reward and global trajectories only during training.

denote the global observable state available to the centralized critic during training. The critic is not given the true C^{out} ; it receives only the same estimate available to the actors.

To obtain a bounded continuous price without post-sampling projection, the actor outputs positive Beta-distribution parameters $\alpha_i(o_i) > 0$ and $\beta_i(o_i) > 0$. It samples

$$y_i(t) \sim \text{Beta}(\alpha_i(o_i(t)), \beta_i(o_i(t))), \quad (49)$$

$$p_i(t) = p_i^{\min} + (p_i^{\max} - p_i^{\min}) y_i(t).$$

The affine transformation keeps every action in $\mathcal{A}_i$ while preserving a well-defined policy likelihood for the PPO update. Exploration uses sampled prices during training. For deterministic evaluation in the case study, hub i uses the mean of its learned Beta policy:

$$p_i^{\text{eval}}(t) = p_i^{\min} + (p_i^{\max} - p_i^{\min}) \frac{\alpha_i(o_i(t))}{\alpha_i(o_i(t)) + \beta_i(o_i(t))}. \quad (50)$$

All actors receive the common reward $r^{\text{sys}}(t)$ from Eq. (39). A single centralized value function $V_{\omega}(s^g(t))$ therefore estimates the expected discounted system return. The system-level temporal-difference error is

$$\delta^{\text{sys}}(t) = r^{\text{sys}}(t) + \gamma V_{\omega}(s^g(t+1)) - V_{\omega}(s^g(t)), \quad (51)$$

and the generalized advantage estimate is

$$\hat{A}^{\text{sys}}(t) = \sum_{\ell=0}^{H-1} (\gamma \lambda_{\text{GAE}})^{\ell} \delta^{\text{sys}}(t+\ell), \quad (52)$$

where H is the rollout horizon and λ_{GAE} is the GAE parameter. The same advantage is used to update every actor because every hub optimizes the same welfare objective.

For actor i , define the probability ratio

$$\varrho_i(t) = \frac{\pi_{\theta_i}(p_i(t) | o_i(t))}{\pi_{\theta_i^{\text{old}}}(p_i(t) | o_i(t))}. \quad (53)$$

The joint actor objective to be maximized is

$$\mathcal{L}_{\text{actor}}(\theta) = \frac{1}{I} \sum_{i \in \mathcal{I}} \mathbb{E}_t \left[\min(\varrho_i(t) \hat{A}^{\text{sys}}(t), \text{clip}(\varrho_i(t), 1 - \epsilon, 1 + \epsilon) \hat{A}^{\text{sys}}(t)) + \beta_{\text{ent}} \mathcal{H}(\pi_{\theta_i}(\cdot | o_i(t))) \right]. \quad (54)$$

where ϵ is the PPO clipping parameter and β_{ent} is the entropy coefficient. The centralized critic is trained by

$$\mathcal{L}_V(\omega) = \mathbb{E}_t \left[\left(V_{\omega}(s^g(t)) - \hat{R}^{\text{sys}}(t) \right)^2 \right], \quad (55)$$

where $\hat{R}^{\text{sys}}(t)$ is the empirical discounted system return. During execution, the critic is discarded and each hub requires only its local actor and the shared preference estimate.

C. Shared Reward and Constraint Handling

The reward in Eq. (39) consists of the system-wide welfare $W^t(p(t))$, an aggregate waiting-time penalty, and an aggregate unmet-energy penalty. Hence an actor is not rewarded

for raising its local profit when the resulting demand shift reduces total welfare or causes violations elsewhere in the hub network.

The constraint treatment has three distinct parts. First, the Beta policy in Eq. (49) enforces the price bounds exactly. Second, the embedded dispatch enforces the PV balance, battery dynamics, charge/discharge limits, and grid import limit in Eqs. (12)–(20). These physical constraints are therefore not repeated as reward penalties. Third, waiting excess and unmet energy are penalized during exploration because they are endogenous consequences of the joint prices and provide informative gradients before the policy becomes feasible.

The coefficients ρ^{wait} and ρ^{unmet} should be chosen relative to normalized reward components so that short-term welfare gains do not dominate service feasibility. Training with penalties is an approximation to the shared-constraint GNEP; it does not by itself guarantee an exact GNE. Accordingly, the final policy is evaluated using system welfare, waiting-time violation frequency and magnitude, unmet energy, and its gap from a small-scale centralized welfare oracle.

D. Dual-Layer Training Procedure

The lower inference update and upper MAPPO update use the same interaction trajectory but operate at different frequencies. Within each period, the current preference estimate is used to predict demand and is included in the actor observations. The true user population then produces the observed demand, after which the lower estimate is updated. Actor and critic parameters are updated only after a rollout has been collected.

For compact notation, let

$$\begin{aligned} \mathbf{o}(t) &= [o_1(t), \dots, o_I(t)], \\ \xi(t) &= (s^g(t), \mathbf{o}(t), \mathbf{p}(t), r^{\text{sys}}(t), s^g(t+1)) \end{aligned} \quad (56)$$

denote the joint transition stored in the rollout buffer.

The critic may use all observable training information but not the hidden true outside-option cost. At deployment, each hub runs its actor independently and may use the deterministic price in Eq. (50), while the lower layer continues to update the shared preference estimate from new aggregate demand observations. The resulting policies provide an adaptive, decentralized approximation to the welfare-optimal GNE defined in Eq. (37).

Algorithm 1 Dual-Layer MAPPO for Welfare-Aligned EV Charging Pricing

- 1: Initialize actors $\{\pi_{\theta_i}\}_{i \in \mathcal{I}}$ and critic V_ω
 - 2: Initialize $\hat{C}^{\text{out}}(0) \in [C^{\text{min}}, C^{\text{max}}]$ and rollout buffer $\mathcal{D}$
 - 3: **for** episode = 1, ..., N_{ep} **do**
 - 4: Initialize traffic, demand, PV, battery, grid-price, and queue states
 - 5: **for** $t = 0, \dots, T - 1$ **do**
 - 6: **for** each hub $i \in \mathcal{I}$ **do**
 - 7: Observe $o_i(t)$ including $\hat{C}^{\text{out}}(t)$
 - 8: Sample $y_i(t)$ and set $p_i(t)$ using Eq. (49)
 - 9: **end for**
 - 10: Form and execute the simultaneous joint price $\mathbf{p}(t)$
 - 11: Compute $\hat{P}_{k,i}(t)$ and $\hat{Q}_i(t)$ using the current preference estimate
 - 12: Observe $Q_i^{\text{obs}}(t)$ generated by the user population; the true C^{out} remains hidden from the learners
 - 13: Compute $\mathcal{L}_{\text{inf}}(t)$ using Eq. (43)
 - 14: Compute $T_i^{\text{wait}}(t)$ and embedded dispatch $e_i^*(t)$ from the observed charging requests
 - 15: Compute $\Pi_i(t)$ and the shared reward $r^{\text{sys}}(t)$
 - 16: Update $\hat{C}^{\text{out}}(t+1)$ using Eq. (47)
 - 17: Advance the exogenous states and construct $\mathbf{o}(t+1)$ and $s^g(t+1)$
 - 18: Store $\xi(t)$ and the actor log-probabilities in $\mathcal{D}$
 - 19: **end for**
 - 20: Compute $\hat{R}^{\text{sys}}$ and $\hat{A}^{\text{sys}}$ from $\mathcal{D}$
 - 21: Update all actors using Eq. (54)
 - 22: Update the centralized critic using Eq. (55)
 - 23: Clear $\mathcal{D}$
 - 24: **end for**
 - 25: **return** decentralized policies $\{\pi_{\theta_i}^*\}_{i \in \mathcal{I}}$ and $\hat{C}^{\text{out}}$
-